MagicCode Super-Resolution Software Specification v1.1.0

1. Document Overview

1.1 Purpose

This document formally defines the technical specifications, functional capabilities, system requirements, and performance benchmarks of the MagicCode Super-Resolution Software (hereinafter referred to as "MagicCode"). It serves as an authoritative reference for the software's development, testing, quality assurance, deployment, and user documentation, ensuring that all stakeholders have a consistent and clear understanding of the product's design objectives and functional boundaries.

1.2 Scope

MagicCode is a specialized image super-resolution software component designed to enhance the resolution of digital images with flexible scaling factors while maintaining or optimizing image detail. The scope of this specification encompasses supported operating systems, hardware platforms, runtime backends, game engine integration plugins, super-resolution functional parameters, input resolution constraints, multi-threading optimization mechanisms, and conformance requirements.

1.3 Definitions, Acronyms and Abbreviations

Term/Acronym	Definition
Super-Resolution (SR)	A digital image processing technology that increases the spatial resolution of an image, generating high-resolution (HR) images from low-resolution (LR) inputs
x86_64	A 64-bit extension of the x86 instruction set architecture (ISA), widely used in desktop and laptop

	computers
ARM64	A 64-bit version of the ARM ISA, commonly adopted in mobile devices, tablets and some laptops
SIMD	Single Instruction, Multiple Data, a parallel processing technology that enables a single instruction to operate on multiple data elements simultaneously
NEON	An advanced SIMD architecture extension for the ARM ISA, enhancing multimedia and signal processing performance
AVX2	Advanced Vector Extensions 2, an extension of the x86 instruction set that expands SIMD capabilities for high-performance computing
4K	A video resolution standard typically referring to 3840×2160 pixels
fps	Frames Per Second, a unit measuring the number of image frames displayed or processed per second

1.4 Document Status

Version	Date	Author	Status	Change Description
V1.1.0	2026-07-02	Technical Documentatio	Updated	Updated backend coverage,

		n Team		scaling range, resolution limits, public modes, and game engine plugin support.

2. System Requirements

2.1 Operating System Compatibility

MagicCode supports multiple mainstream operating systems across desktop and mobile platforms, with specific version requirements as follows:

- Windows: Supports Windows 10 (64-bit) and above
- Android: Supports Android 8.0 and above
- iOS: Supports iOS 13.0 and above
- iPadOS: Supports iPadOS 13.0 and above
- macOS: Supports macOS 12.0 (Monterey) and above

2.2 Hardware Requirements

2.2.1 CPU Architecture Support

The software is compatible with the following 64-bit CPU architectures to ensure broad hardware adaptability:

- x86_64: Applicable to mainstream Intel and AMD desktop/laptop CPUs
- ARM64: Applicable to Apple Silicon (e.g., M-series chips) and mainstream ARM-based mobile CPUs

2.2.2 SIMD Instruction Set Support

To optimize image processing performance, MagicCode leverages advanced SIMD instruction sets for parallel computing, with mandatory support for the following instruction sets based on CPU architecture:

- NEON: Mandatory for ARM64 architecture CPUs, enabling efficient multimedia data processing on mobile and Apple Silicon devices
- AVX2: Mandatory for x86_64 architecture CPUs, providing enhanced vector processing capabilities for desktop platforms

2.3 Runtime Backend Support

MagicCode supports the following runtime backends: x86, NEON, Metal, OpenGL, Vulkan, and OpenGLES.

x86 and NEON provide CPU execution paths. Metal, OpenGL, Vulkan, and OpenGLES provide GPU execution paths for Apple, Windows, and Android platforms as applicable.

2.4 Game Engine and Plugin Integration Support

Android supports Unreal Engine plugins for UE5.0 through UE5.8.

iOS supports Unreal Engine plugins for UE5.6 through UE5.8.

Android and iOS support Unity integration plugins.

3. Functional Specifications

3.1 Super-Resolution Scaling Ratios

MagicCode supports flexible super-resolution scaling factors in the inclusive range [1, 8], including integer and non-integer ratios where supported by the selected backend and model configuration.

- Supported scaling range: any requested scale from 1.0x to 8.0x, subject to backend and model capability.
- We recommend using 2x super-resolution, as it fully leverages the efficiency and high quality of the PACA algorithm.
- Output width and height are computed from the requested floating-point scaling factor and rounded to the nearest pixel dimension.

3.2 Super-Resolution Modes

Two public super-resolution processing modes are provided to meet different application scenarios. Internal implementation modes may be selected automatically by the library and are not exposed through the public API.

Mode	Characteristics	Typical Use Case
Highspeed	Prioritizes throughput and low latency, suitable for real-time rendering, video playback, and interactive applications	Real-time or latency-sensitive super-resolution workloads

Speed	Balances processing speed and output quality, suitable for higher-quality image/video enhancement when more processing budget is available	Quality-oriented super-resolution workloads

3.3 Input Resolution Constraints

To ensure processing stability and efficiency, MagicCode defines clear constraints on the input image resolution (width × height):

- Maximum input resolution: 4032×4032 pixels
- Minimum input resolution: 64×64 pixels

If the input image resolution exceeds 4032×4032 pixels or is lower than 64×64 pixels, the software will return an error prompt and refuse to process. Users need to adjust the input resolution to the valid range first.

4. Performance Optimization Specifications

4.1 Multi-Threading Processing Mechanism

MagicCode adopts a dynamic multi-threading strategy to improve processing efficiency for large-size images. The activation conditions and constraints of multi-threading are as follows:

- Activation condition: The number of input image rows (height) is ≥ 256. For images with fewer than 256 rows, single-threaded processing is used to avoid thread scheduling overhead.
- Maximum thread count: 8 threads. The software automatically adjusts the actual number of active threads based on the current CPU core count and system resource usage, but will not exceed 8 threads to prevent resource contention.

On multi-core CPU platforms, this mechanism can significantly reduce the total processing time for high-resolution images, especially when processing images up to the supported maximum resolution.

5. Conformance Criteria

All functional and performance indicators specified in this document shall be verified through formal testing. The software shall be deemed conformant only if it meets 100% of the mandatory requirements outlined herein. Any deviations from the specifications must be approved by the product management and technical steering committees, with corresponding revisions to this document.

6. References

1. ARM Architecture Reference Manual: NEON Extension
2. Intel 64 and IA-32 Architectures Software Developer's Manual: AVX2 Instructions
3. ISO/IEC 29100: Information technology — Security techniques — Privacy framework